providing there is a match. For example, an investor sending a market order to a "grey pool" may transact at a price better than the NBBO if there is a hidden order in the venue at a better price.

Dark Pool Controversies

Historically there has been a large amount of debate surrounding dark pool executions, adverse selection, and toxic order flow. Adverse selection refers to situations when you use a dark pool and have the order executed fully (100%). Subsequent price movement is in your favor (e.g., buys become cheaper and sells become higher) so you would have been better off waiting to make the trade. And when you do not execute in the dark pool or execute less than the full order (<100%) the subsequent price movement is away from your order (e.g., buys become more expensive and sells become lower in value). The belief is that there is either some information leakage occurring in the dark pool or the interaction with high frequency orders is toxic, meaning that the high frequency traders are able to learn information about the order, such as the urgency of the investor or the number of shares that still need to be executed. In turn, they adjust their prices based on leaked knowledge. However, we have not yet found evidence of adverse selection in dark pools to confirm these suspicions.

But let us evaluate the above situation from the order level. Suppose we have a buy order for 100,000 shares and there is a seller with an order for 200,000 shares. Thus, there is a sell imbalance of −100,000 shares. If both parties enter the order into the crossing network (dark pool or other type of matching system) there will be 100,000 shares matched with the buy order being 100% filled and the sell order being only 50% filled. The seller will then need to transact another 100,000 shares in the market and the incrementing selling pressure will likely push the price down further due to the market impact cost of their order. So the downward price movement is caused by the market imbalance, not by the dark pool. Next, suppose that the seller only has 50,000 shares. Thus, there is a +50,000 buy imbalance. If these orders are entered into the crossing network, 50,000 shares of the buy order will match. The buyer will then need to transact the additional 50,000 shares in the displayed market where the buying pressure will likely push the price up further. Thus we can see that the adverse price movement is caused by the market imbalance and not the dark pool. This type of price movement is commonly observed in times of market imbalances.

To be fair, there was a time when dark pools and venues allowed flash orders to be entered into their systems. These flash orders would provide